ModelIt!

Student Activity Pack

Can We Stop an Earthquake from Breaking Buildings?

Engineering Solutions to Protect People from Natural Hazards

Grade 4 | NGSS 4-ESS3-2 | Cause and Effect

STUDENT NAME _____ DATE _____

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1 Pre-Assessment

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What Do You Know? Answer each question below to the best of your ability. There are no wrong answers — this helps your teacher understand what you already know!

1. **Have you ever felt an earthquake or seen earthquake damage on the news?**

2. **Why do some buildings fall down during earthquakes and others stay standing?**

3. **If you were going to build an earthquake-proof building, what would you do differently?**

4. **Draw what you think happens to a building during an earthquake.**

2 Scoring Rubric

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Your teacher will use this rubric to assess your work. Read through each level to understand what is expected!

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

3 Component Analysis

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What is a Component?

A **component** is a part of a system that plays a specific role. Components can be **EXTERNAL** (come from outside the system) or **INTERNAL** (happen within the system).

Sort These Components

Earthquake Strength | Building Flexibility | Damage Level

EXTERNAL	INTERNAL

Explain your reasoning: Why did you sort each component the way you did?

4 Model Recording Page

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Draw your model showing all components and their relationships. Use the relationship key below to label your arrows!

+ **POSITIVE:** When A increases, B also increases

− **NEGATIVE:** When A increases, B decreases

Draw your model here

? Explain ONE relationship in your model. What happens when one component changes?

5 Simulation Observations

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Run each scenario in ModelIt! First, write your **prediction**. Then run the simulation and record what you **observe**.

1 Scenario 1: Small Tremor

Low earthquake, moderate flexibility

MY PREDICTION

WHAT I OBSERVED

2 Scenario 2: Major Quake, Stiff Building

High earthquake, low flexibility

MY PREDICTION

WHAT I OBSERVED

3 Scenario 3: Major Quake, Flexible Building

High earthquake, high flexibility

MY PREDICTION

WHAT I OBSERVED

6 Research & Extend

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> Real systems are more complex than our model! Here are additional components scientists study. Choose one and explain how it could connect to our model.

Possible New Components

Foundation Depth — Deeper foundations anchor buildings more securely to stable ground below the shaking surface

Building Height — Taller buildings sway more during earthquakes and need special engineering to stay safe

Ground Type — Buildings on soft soil shake more than buildings on hard rock during the same earthquake

My Extension

A Which new component would you add to the model?

B How would it connect to the existing components? (Draw or describe)

C What evidence or reasoning supports your connection?

7 Reflection Questions

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The Big Question

How do engineers design buildings that can survive an earthquake without falling down? Use evidence from your simulation to support your answers.

1 Why is a flexible building safer than a stiff building during an earthquake?

2 How is earthquake engineering similar to how trees survive strong windstorms?

3 If we cannot stop earthquakes from happening, what CAN we do to keep people safe?

4 What other natural hazards could engineers design solutions for?

5 How does your model help engineers make better decisions about building design?

8 STEM Challenge: Design an Earthquake-Proof Structure

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YOUR ENGINEERING MISSION

Design and build a model structure that can survive simulated earthquake shaking on a shake table without collapsing.

THE SCENARIO

A city in earthquake country needs new buildings designed to protect people during the next big quake. Your engineering team must design a structure that stays standing when the ground shakes. You will test your design on a shake table and compare results with other teams.

Engineering Design Thinking

1. WHAT MATERIALS WILL YOU USE TO MAKE YOUR STRUCTURE FLEXIBLE BUT STRONG?

2. HOW WILL YOU CONNECT THE BASE OF YOUR STRUCTURE TO THE GROUND — RIGIDLY OR WITH SOME MOVEMENT ALLOWED?

3. WHAT SHAPE WILL YOUR STRUCTURE BE — TALL AND THIN, OR WIDE AND LOW?

4. HOW WILL YOU MAKE SURE YOUR STRUCTURE CAN SURVIVE MULTIPLE SHAKES, NOT JUST ONE?

My Design

DESIGN AREA — DRAW YOUR SOLUTION

N

! Explain your design using scientific evidence from the simulation:

Real Career Connection

Structural Engineers and Seismic Design Specialists design buildings, bridges, and structures that can withstand earthquakes, hurricanes, and other natural forces. They earn $70,000–$130,000/year.